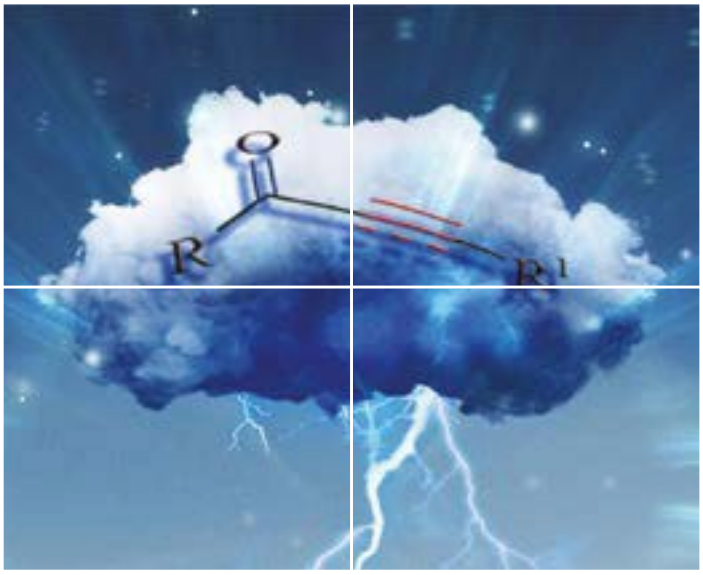
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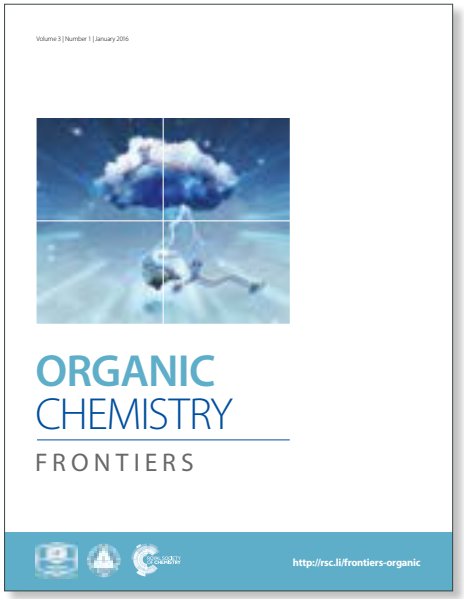
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ARTICLE

2,3-Allenoic acids *via* palladium-catalyzed carboxylation of propargylic alcohols

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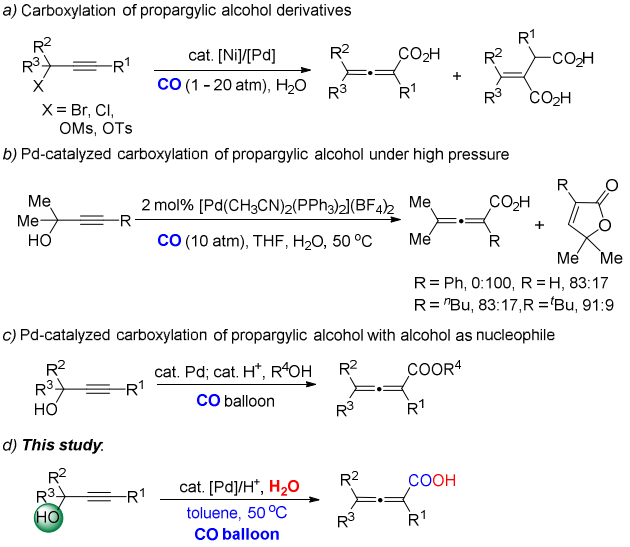
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Wei-Feng Zheng,[a] Wanli Zhang,[c] Jianhao Huang,[d] Yibo Yu,[d] Hui Qian,*[a] Shengming Ma*[a,b]

A straightforward protocol for the synthesis of *tetra*-substituted 2,3-allenoic acids via a catalytic carboxylation of the readily available propargylic alcohols in the presence of CO using palladium and Brønsted acid as the cocatalysts has been established. This atom-economic reaction proceeded under mild conditions to afford 2,3-allenoic acids with a broad scope tolerating useful functional groups.

Introduction

Allenes are not only a common structural motif in natural products, biologically active compounds,[1] and advanced materials,[2] but also serve as attractive precursors in a variety of organic transformations.[3] Among functionalized allenes, 2,3-allenoic acids are of great interest since they may be readily transformed into allenols (then, amines, halides, or tosylates, etc.),[4] allenals,[5] allenoates,[6] allenamides,[7] and butenolides.[8] Thus, the development of efficient methods for the synthesis of these compounds is of great interest. Traditional approach to 2,3-allenoic acids is the hydrolysis of 2,3-allenoates, which usually yielded a mixture of allenoic acids and alkynoic acids.[9,10] Carboxylation reaction of allenylmetallic reagents with carbon dioxide,[11] the oxidation of allenols,[12] and the isomerization of but-3-ynoic acid and 4-(trimethylsilyl)but-3-ynoic acid were also reported,[11d,13] but these methods suffer from poor atom- or step-economy, harsh reaction conditions, or limited substrate scope. It is believed that the carboxylation of the readily available propargylic alcohols or their derivatives such as carbonates, acetates, etc. with CO in the presence of water should be the most atom economic way for synthesis of allenoic acids. The reports on such a strategy are very limited although the related carbonylations of the related propargylic derivatives affording the allenoates are well established.[10] The challenge should be the poor nucleophilicity of water. Nickel- or palladium-catalyzed carboxylations of propargylic halides or mesylates were reported to afford a mixture of 2,3-allenoic acids and itaconic acids (Scheme 1a).[10c,14] In 1994, Inoue and co-workers developed the Pd-catalyzed carboxylation of 2-methyl-3-alkynols affording a mixture of 2,3-allenoic acids and 2(5*H*)-furanones with a high pressure of CO (10 atm) (Scheme 1b).[15] It should also be noted that the reaction of propargylic alcohol in the presence of a transition metal catalyst and carbon monoxide afforded lactones.[15,16] Previously, we reported the alkoxycarbonylation of propargylic alcohols (Scheme 1c).[10l] Here, we report the first example on palladium-catalyzed carboxylation of propargylic alcohols with a catalytic phosphoric acid as the cocatalyst to afford *tetra*-substituted 2,3-allenoic acids exclusively.



Scheme 1. Transition metal-catalyzed carboxylation approaches toward 2,3-allenoic acids

Results and discussion

Our investigations commenced with the Pd/H$^+$-cocatalyzed reaction of 2-phenyloct-3-yn-2-ol (**1a**) as model substrate. In the presence of [PdCl(π-allyl)]$_2$, DPEphos, (PhO)$_2$POOH, and

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†Dedicated to Professor Xiyan Lu on the occasion of his 90th birthday.

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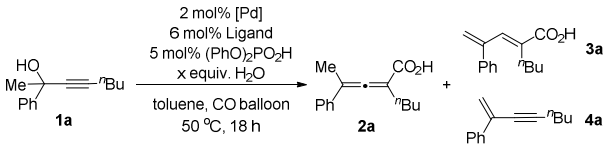
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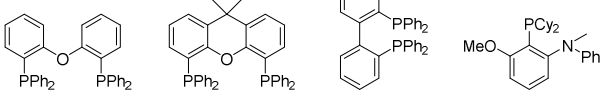
H_2O in toluene with CO (provided by a balloon), the reaction furnished the expected 2,3-allenoic acid **2a** in 62% yield, albeit together with 4% yield of 2,4-alkadienoic acid **3a**[17] (Table 1, entry 1).[10l] It was noted that the yield was much lower with Xantphos as the ligand (Table 1, entry 2). When BIPHEP was employed, a mixture of allenoic acid **2a**, 2,4-dienoic acid **3a,** and enyne **4a** were obtained with a poor selectivity (Table 1, entry 3). When PPh_3 was employed, this reaction afforded the dehydration product enyne **4a** as the major product (Table 1, entry 4). Zheda-phos[18] exhibited no catalytic activity (Table 1, entry 5). Catalyst screening was conducted with DPEphos as the ligand: No expected product was formed with $Pd(PPh_3)_4$ or $Pd_2(dba)_3{\cdot}CHCl_3$. Interestingly, it turned out that $PdCl_2$ is the best (Table 1, entries 6, 7, and 8). The efficiency was reduced when the amount of water was decreased to 4 equiv (Table 1, entry 9). Very limited level of reaction was observed in the absence of water (Table 1, entry 10). Thus, optimal results were defined as 2 mol% of $PdCl_2$, 6 mol% DPEphos, 5 mol% $(PhO)_2POOH$, and 8.0 equiv of H_2O in toluene at 50 °C under 1 atm of CO (Table 1, entry 8) affording the desired allenoic acid product in 97% NMR yield.

Table 1. Optimization of the reaction conditions for the carboxylation of 1a[a]



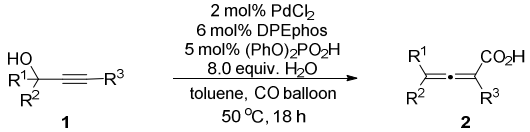
entry	[Pd]	Ligand	x equiv of H_2O	NMR yield[b] (%)			Recovery of **1a**[b] (%)
				2a	**3a**	**4a**	
1[c]	$[PdCl(\pi\text{-allyl})]_2$	DPEphos	8	62	4	/	16
2	$[PdCl(\pi\text{-allyl})]_2$	Xantphos	8	3	/	1	96
3	$[PdCl(\pi\text{-allyl})]_2$	BIPHEP	8	20	3	3	72
4	$[PdCl(\pi\text{-allyl})]_2$	PPh_3	8	8	/	10	82
5	$[PdCl(\pi\text{-allyl})]_2$	Zheda-phos	8	/	/	3	89
6	$Pd(PPh_3)_4$	DPEphos	8	/	/	5	67
7	$Pd_2(dba)_3{\cdot}CHCl_3$	DPEphos	8	/	3	7	24
8	**$PdCl_2$**	**DPEphos**	**8**	**97**	**2**	**/**	**/**
9	$PdCl_2$	DPEphos	4	84	1	3	11
10	$PdCl_2$	DPEphos	0	14	/	8	78

[a] Reaction conditions: **1a** (0.5 mmol), [Pd] (2 mol%), Ligand (6 mol%), $(PhO)_2PO_2H$ (5 mol%),and H_2O (x equiv.) in toluene (2.5 mL) at 50 °C with a CO balloon unless otherwise noted. [b] Determined by ^1H NMR analysis using dibromomethane as the internal standard. [c] The reaction was carried out with 1 mmol of **1a** in 5.0 mL toluene.



With the optimal reaction conditions in hand, the scope of the method was examined. As shown in Table 2, a variety of tertiary propargylic alcohols bearing differently substituted phenyl group (R^1) were initially surveyed. The substrates with electron-donating and electron-withdrawing group substituted aryl group could be employed in our protocol (**2a-2f**). There is no obvious influence on the steric effect since substrates bearing the methyl group at the *o*-, *m*-, and *p*-position of the phenyl group all afford the desired products with high yields (**2b**-**2d**). The C-Br bond survived affording **2f**. R^1 may also be an alkyl group (entry 17). The reaction of 2-(naphth-2-yl)oct-3-yn-2-ol delivered **2g** in 93% yield. When R^1 is a phenyl ring, R^2 may vary from alkyl to phenyl group giving the carboxylation products **2h-2j** in 53-97% yields. Next, we turned our attention to the scope of the R^3 group: alkyl group such as *n*-propyl, *n*-hexyl, *i*-propyl, chloropropyl and phenyl may all work to afford the corresponding products in good to excellent yields (**2k**-**2p**). The structure of **2p** was confirmed by single-crystal X-ray diffraction analysis (Figure 1).[19] In addition, all alkyl substitution propargylic alcohol **1q** was also amenable to carboxylation to afford **2q** in 56% yield.

Table 2. Substrate Scope of propargylic alcohols[a]



entry	**1**				isolated yield of **2** (%)[b]	
	R^1	R^2	R^3			
1	Ph	Me	^{n}Bu	**(1a)**	92	**(2a)**
2	2-MeC_6H_4	Me	^{n}Bu	**(1b)**	87 (7)	**(2b)**
3	3-MeC_6H_4	Me	^{n}Bu	**(1c)**	93	**(2c)**
4	4-MeC_6H_4	Me	^{n}Bu	**(1d)**	82 (7)	**(2d)**
5	3-$MeOC_6H_4$	Me	^{n}Bu	**(1e)**	71	**(2e)**
6	4-BrC_6H_4	Me	^{n}Bu	**(1f)**	84 (4)	**(2f)**
7	2-Naphthyl	Me	^{n}Bu	**(1g)**	93	**(2g)**
8	Ph	^{n}Pr	^{n}Bu	**(1h)**	97	**(2h)**
9	Ph	Bn	^{n}Bu	**(1i)**	74 (18)	**(2i)**
10	Ph	Ph	^{n}Bu	**(1j)**	48 (53)	**(2j)**
11	Ph	Et	^{n}Pr	**(1k)**	87	**(2k)**
12	Ph	Et	^{n}Hex	**(1l)**	96	**(2l)**
13	Ph	Et	^{i}Pr	**(1m)**	79 (12)	**(2m)**
14	Ph	Et	-$(CH_2)_3Cl$	**(1n)**	96	**(2n)**
15	Ph	^{n}Bu	^{n}Hex	**(1o)**	93	**(2o)**
16[c]	Ph	Me	Ph	**(1p)**	51 (34)	**(2p)**
17	^{n}Pr	Me	^{n}Bu	**(1q)**	56 (34)	**(2q)**

[a] Reaction conditions: **1** (1.0 mmol), $PdCl_2$ (2 mol%), DPEphos (6 mol%), $(PhO)_2PO_2H$ (5 mol%), and H_2O (8 equiv.) in toluene (5.0 mL) at 50 °C under 1 atm of CO unless otherwise noted. [b] Recovery of starting material determined by ^1H NMR analysis using dibromomethane as internal standard is shown in parentheses. [c] The reaction was carried out in 36 h.

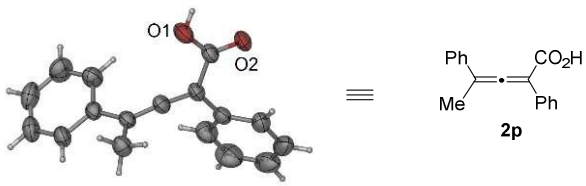


Figure 1. ORTEP representation of 2p

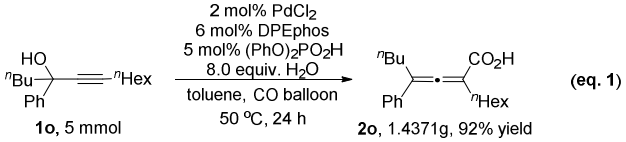
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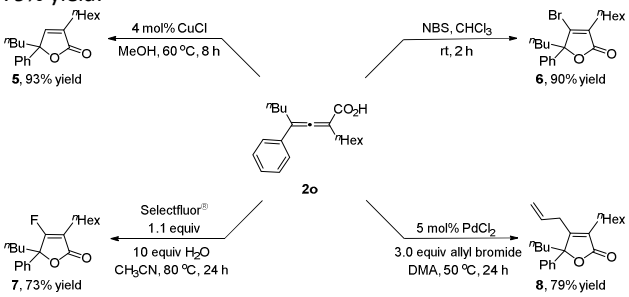
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Gram-scale reaction of **1o** afforded carboxylated product **2o** in 92% yield under the standard conditions (eq. 1).



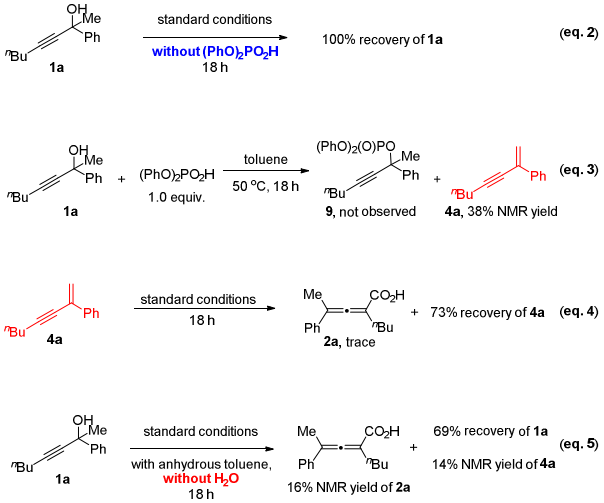
(eq. 1)

The synthetic application of product **2o** is demonstrated in Scheme 2: When **2o** was treated with CuCl, the cycloisomerization product 2(5*H*)-furanone **5** was obtained in 93% yield.[20] 4-Bromo-2(5*H*)-furanone **6** was obtained in excellent yield via a bromolactonization reaction. This 2,3-allenoic acid **2o** may also undergo a fluorolactonization reaction with Selectfluor to afford **7** in 73% yield.[21] A palladium-catalyzed cyclization-allylation reaction of **2o** in the presence of allyl bromide furnished allylated butenolide **8** with 79% yield.[8g]

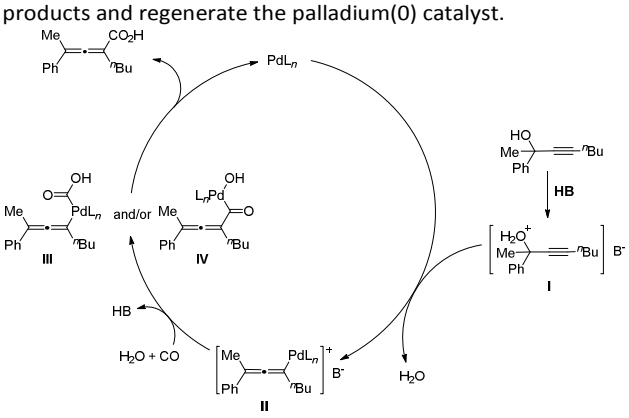


Scheme 2. Cyclization reactions 2,3-allenoic acid 2o

To gain further insight into the mechanism, we carried out some control experiments. Firstly, no reaction occurred without the addition of $(PhO)_2PO_2H$ (eq. 2). We hypothesize the phosphate **9** may be formed in situ as the reactive species. Thus, we mixed **1a** with 1 equiv. $(PhO)_2PO_2H$ at 50 °C, however, the formation of **9** was not observed (eq. 3), instead, the dehydration product enyne **4a** was formed. When enyne **4a** was submitted to the standard conditions, only trace amount of **2a** was detected, which suggests the enyne is not the possible intermediate (eq. 4). The hydroxy group in the carboxyl unit of the allenoic acids may come from the hydroxy group in the propargylic alcohols, however, the reaction rate was very slow in the absence of water (eq. 5), which suggested the hydroxy group mostly came from the added water.



A possible catalytic cycle for this Pd/H^+ co-catalyzed carboxylation reaction was proposed (Scheme 3). At the beginning, Brønsted acid activated propargylic alcohol to generate the oxonium intermediate **I**, which would undergo oxidative addition with Pd(0) to produce the key allenylpalladium intermediate **II**. The allenylpalladium species would react with CO and H_2O to give intermediate **III** and/or **IV**. Subsequent reductive elimination would furnish the products and regenerate the palladium(0) catalyst.



Scheme 3. A proposed catalytic cycle

Conclusions

In summary, we have developed a straightforward protocol for the synthesis of *tetra*-substituted 2,3-allenoic acids via a catalytic carboxylation of the readily available propargylic alcohols in the presence of CO using palladium and Brønsted acid as the cocatalysts. The reaction proceeded in an atom-economical way under mild conditions. Further studies related to this topic are currently ongoing in our laboratory.

Conflicts of interest

There are no conflicts to declare.

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Notes and references

1 A. Hoffmann-Röder and N. Krause, *Angew. Chem., Int. Ed.*, 2004, **43**, 1196.
2 P. Rivera-Fuentes, and F. Diederich, *Angew. Chem., Int. Ed.*, 2012, **51**, 2818.
3 For selected reviews, see: (a) N. Krause and A. S. K. Hashimi, ed. *Modern Allene Chemistry*, Wiley-VCH, Weinheim, Germany, 2004; (b) R. Zimmer, C. U. Dinesh, E. Nandanan and F. A. Khan, *Chem. Rev.*, 2000, **100**, 3067; (c) S. Ma, *Acc. Chem. Res.*, 2003, **36**, 701; (d) S. Ma, *Chem. Rev.*, 2005, **105**, 2829; (e) Ma, S. *Aldrichimica Acta*, *2007*, **40**, 91; (f) S. Ma, *Acc. Chem. Res.*, 2009, **42**, 1679; (g) C. Aubert, L.

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Fensterbank, P. Garcia, M. Malacria, and A. Simonneau, *Chem. Rev.*, 2011, **111**, 1954; (h) S. Yu and S. Ma, *Angew. Chem., Int. Ed.*, 2012, **51**, 3074; (i) J. Ye and S. Ma, *Acc. Chem. Res.*, 2014, **47**, 989; (j) B. Alcaide, P. Almendros and C. Aragoncillo, *Chem. Soc. Rev.*, 2014, **43**, 3106; (k) M. P. Muñoz, *Chem. Soc. Rev.*, 2014, **43**, 3164.

4 (a) S. Ma, and S. Zhao, *J. Am. Chem. Soc.*, 1999**, 121***,* 7943; (b) S. Ma and S. Zhao, *J. Am. Chem. Soc.*, 2001**, 123***,* 5578; (c) S. Ma and W. Gao, *J. Org. Chem.*, 2002, **67**, 6104; (d) C. Fu, J. Li and S. Ma, *Chem. Commun.*, 2005, **0**, 4119; (e) Z. Lu and S. Ma, *Adv. Synth. Catal.*, 2007, **349**, 1225; (f) Y. Deng, Y. Yu and S. Ma, *J. Org. Chem.*, 2008, **73**, 585; (g) Y. Deng, X. Jin, C. Fu and S. Ma, *Org. Lett.*, 2009, **11**, 2169; (h) W. Kong, C. Fu and Ma, S. *Chem. Commun.*, 2009, **0**, 4572. (i) J. Li, W. Kong, Y. Yu, C. Fu and S. Ma, *J. Org. Chem.*, 2009, **74**, 8733; (j) S. Li, B. Miao, W. Yuan and S. Ma, *Org. Lett.*, 2013, **15**, 977; (k) B. Alcaide, P. Almendros, A. Luna and E. Soriano, *J. Org. Chem.*, 2015, **80**, 7050.

5 (a) D. Ma, Y. Qiu, J. Dai, C. Fu and S. Ma, *Org. Lett.*, 2014, **16**, 4742; (b) Y. Qiao, D. Wei and J. Chang, *J. Org. Chem.*, 2015, **80***,* 8619; (c) J. Dai, M. Wang, G. Chai, C. Fu and S. Ma, *J. Am. Chem. Soc.*, 2016, **138**, 2532.

6 (a) X. Jiang and S. Ma, *J. Am. Chem. Soc.*, 2007, **129**, 11600; (b) Y. Shi, K. E. Roth, S. D. Ramgren and S. A. Blum, *J. Am. Chem. Soc.*, 2009, 131, 18022; (c) G. Chai, Z. Lu, C. Fu and S. Ma, *Adv. Synth. Catal.*, 2009, **351**, 1946; (d) S. Y. Lee, Y. Fujiwara, A. Nishiguchi, M. Kalek and G. C. Fu, *J. Am. Chem. Soc.*, 2015, **137**, 4587; (e) W. Yao, X. Dou, S. Wen, J. Wu, J. J. Vittal and Y. Lu, *Nat. Commun.*, 2016, **7**, 13024.

7 (a) L. S. Trifonov and A. S. Orahovats, *Helv. Chim. Acta.*, 1986, **69**, 1585; (b) L. S. Trifonov and A. S. Orahovats, *Helv. Chim. Acta.*, 1987, **70**, 262; (c) S. Ma and H. Xie, *Tetrahedron*, 2005, **61**, 251; (d) S. Ma and H. Xie, *J. Org. Chem.*, 2002, **67***,* 6575; (e) G. Chen, C. Fu and S. Ma, *Org. Lett.*, 2009, **11**, 2900.

8 (a) S. Ma and Z. Shi, *J. Org. Chem.*, 1998, **63***,* 6387; (b) S. Ma and S. Wu, *J. Org. Chem.*, 1999, **64***,* 9314; (c) S. Ma and Z. Yu, *Angew. Chem., Int. Ed.* 2002, **41**, 1775; (d) S. Ma and Z. Shi, *Chem. Commun.*, 2002, **0**, 540; (e) S. Ma and Z. Yu, *Angew. Chem., Int. Ed.*, 2003, **42**, 1955; (f) S. Ma and Z. Yu, *Org. Lett.*, 2003, **5**, 1507; (g) S. Ma and Z. Yu, *J. Org. Chem.*, 2003, **68**, 6149; (h) S. Ma, F. Pan, X. Hao and X. Huang, *Synlett*, 2004, **1**, 85; (i) S. Ma and Z. Gu, *J. Am. Chem. Soc.*, 2005, **127**, 6182; (j) S. Ma, Z. Gu and Y. Deng, *Chem. Commun.*, 2006, **0**, 94; (k) Z. Gu and S. Ma, *Angew. Chem., Int. Ed.*, 2006, **45**, 6002; (l) Z. Gu, X. Wang, W. Shu and S. Ma, *J. Am. Chem. Soc.*, 2007, **129**, 10948; (m) G. Chen, R. Zeng, Z. Gu, C. Fu and S. Ma, *Org. Lett.*, 2008, **10**, 4235; (n) Q. Yu and S. Ma, *Chem. Eur. J.*, 2013, **19**, 13304; (o) R. Lü, X. Cheng, X. Zheng and S. Ma, *Chem. Commun.*, 2014, **50**, 1537; (p) K. Matsushita, K. Suzuki and K. Ohmori, *Synlett*, 2017, **28**, 944; (q) S. Pan, Y. Huang, X.-H. Xu, F.-L. Qing, *Org. Lett.*, 2017, **19**, 4624.

9 For the synthesis of allenoates *via* Wittig-type reactions, see: (a) S. Ramaswamy, R. A. H. F. Hui and J. B. Jones, *J. Chem. Soc. Chem. Commun.*, 1986, **0**, 1545; (b) Y.-F. Ao, D.-X. Wang, L. Zhao and M.-X. Wang, *J. Org. Chem.*, 2014, **79**, 3103; (c) J. A. Marshall, E. D. Robinson and A. Zapata, *J. Org. Chem.*, 1989, **54**, 5854; (d) C. Li, X. Wang, X. Sun, Y. Tang, J. Zheng, Z. Xu, Y. Zhou and L. Dai, *J. Am. Chem. Soc.*, 2007, **129**, 1494.

10 For the synthesis of allenoates *via* transition metal-catalyzed carboxylation of propargylic alcohols and their derivatives with CO in the presence of an alcohol, see: (a) J. Tsuji, T. Gugiura and I. Minami, *Tetrahedron Lett.*, 1986, **27**, 731; (b) J. A. Marshall and M. A. Wolf, *J. Org. Chem.*, 1996, **61**, 3238; (c) J. A. Marshall, M. A. Wolf, E. M. Wallace, *J. Org. Chem.*, 1997, **62**, 367; (d) G. E. Akpınar, M. Kuş, M. Uçüncü, E. Karakuş and L. Artik, *Org. Lett.*, 2011, **13**, 748; (e) Y. Wang and S. Ma, *Adv. Synth. Catal.*, 2013, **355**, 741; (f) Y. Wang, W. Zhang and S. Ma, *J. Am. Chem. Soc.*, 2013, **135**, 11517; (g) Y. Wang, W. Zhang and S. Ma, *Org. Chem. Front.*, 2014, **1**, 807; (l) W. Zhang, C. Huang, Y. Yuan and S. Ma, *Chem. Commun.* 2017, **53**, 12430.

11 (a) H. Westmijze, H. Kleijn, H. J. T. BOS and P. Vermeer, *J. Organometallic Chem.*, 1980, **199**, 293; (b) J.-C. Clinet and G. Linstrumelle, *Synthesis*, 1981, **11**, 875; (c) N. R. Pearson, G. Hahn and G. Zweifel, *J. Org. Chem.*, 1982, **47**, 3364; (d) B. Miao, G. Li and S. Ma, *Chem. Eur. J.*, 2015, **21**, 17224.

12 E. Ferre, G. Gil, M. Bertrand and J. L. Petit, *Appl Microbiol Biotechnol*, 1985, **21**, 258.

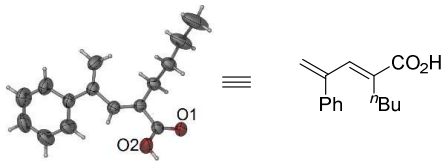
13 (a) G. Eglinton, E. R. H. Jones, G. H. Mansfield and M. C. Whiting, *J. Chem. Soc.*, 1954, **0**, 3197; (b) K. Nogi, T. Fujihara, J. Terao and Y. Tsuji, *Chem. Commun.*, 2014, **50**, 13052.

14 (a) H. Arzoumanian, F. Cochini, D. Nuel, J. F. Petrignani and N. Rosas, *Organometallics*, 1992, **11**, 493; (b) H. Arzoumanian, F. Cochini, D. Nuel and N. Rosas, *Organometallics*, 1993, **12**, 1871; (c) Z. Gu and S. Ma, *Chem. Eur. J.*, 2008, **14**, 2453; (d) G. Nemecek, R. Thomas, H. Goesmann, C. Feldmann and J. Podlech, *Eur. J. Org. Chem.*, 2013, 6420.

15 K. Matsushita, T. Komori, S. Oi and Y. Inoue, *Tetrahedron Lett.*, 1994, **35**, 5889.

16 For selected examples of cyclocarbonylation of propargylic alcohol, see: (a) W.-J. Xiao, and Alper, H. *J. Org. Chem.*, 1997, **62***,* 3422; (b) W.-Y. Yu and H. Alper, *J. Org. Chem.*, 1997, **62***,* 5684; (c) Y. Fukuta, I. Matsuda and K. Itol, *Tetrahedron Lett.* 2001, **42**, 1301; (d) Z.-X. Jiang and Qing, F.-L. *Tetrahedron Lett.*, 2001, **42**, 9051; (e) S. Ma, B. Wu and S. Zhao, *Org. Lett.* 2003, **5**, 4429; (f) C. Consorti, G. Ebeling and J. Dupont, *Tetrahedron Lett.*, 2002, **43**, 753. (g) X. Yin, H. Mohammad, H. E. Eldesouky, A. Abdelkhalek, M. N. Seleem and M. Dai, *Chem. Commun.*, 2017, **53**, 7238.

17 The structure of **3a** was confirmed by single-crystal X-ray diffraction analysis. CCDC-1836594 contains the supplementary crystallographic data for compound **3a**. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. Crystal data for compound **3a**: $C_{15}H_{18}O_2$, MW = 230.29, triclinic, space group *P*-1, final *R* indices [$I > 2\sigma(I)$], R_1 = 0.0680, wR_2 = 0.2031; *R* indices (all data), R_1 = 0.0879, wR_2 = 0.2213, *a* = 5.6491(4) Å, *b* = 11.5037(11) Å, *c* = 11.5175(7) Å, α = 111.795(7)°, β = 99.480(6)°, γ = 90.087(7)°, *V* = 683.87(9) Å^3, *T* = 293(2) K, Z = 2, reflections collected/unique 5976/2405 [R_{int} = 0.0418], no. of observations [> 2 $\sigma(I)$] 1683, parameters: 156.
ORTEP representation of **3a**:



18 P. Li, B. Lü, C. Fu and S. Ma, *Adv. Synth. Catal.*, 2013, **355**, 1255.

19 CCDC-1821146 contains the supplementary crystallographic data for compound **2p**. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. Crystal data for compound **2p**: $C_{17}H_{14}O_2$, MW = 250.28, monoclinic, space group *P*2(1)/*c*, final *R* indices [$I > 2\sigma(I)$], R_1 = 0.0375, wR_2 = 0.1017; *R* indices (all data), R_1 = 0.0413, wR_2 = 0.1049, *a* = 10.5915(2) Å, *b* = 15.1258(2) Å, *c* = 9.0257(2) Å, α = 90.00°, β = 109.468(2)°, γ = 90.00°, *V* = 1363.29(5) Å^3, *T* = 293(2) K, Z = 4, reflections collected/unique 13025/2423 [R_{int} = 0.0410], no. of observations [> 2 $\sigma(I)$] 2174, parameters: 175.

20 S. Ma, Z. Yu and S. Wu, *Tetrahedron*, 2001, **57***,* 1585.

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21 C. Zhou, Z. Ma, Z. Gu, C. Fu and S. Ma, *J. Org. Chem.*, 2008, **73**, 772.

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